# Principal Component Analysis (PCA)

Interpretation, Validation, Outliers, and Process Monitoring

Questions Only

TTK4260: Multivariate Data Analysis and ML

**Question 1** *(Score plot: clusters, gradients, outliers)*

*[6 points]* You run PCA with two components and plot a score plot ($t_1$ vs. $t_2$). You observe: (i) two tight clusters separated mainly along $t_1$, (ii) points within each cluster are smoothly ordered from left to right when colored by a continuous variable "feed rate", and (iii) one isolated point far in the top-right.

(a) What do the two clusters suggest?

(b) What does the smooth ordering with feed rate suggest (often called a "trend" or "gradient")?

(c) List two plausible reasons for the isolated point, and what you would check next.

**Question 2** *(Score line plot: drift vs step vs cycles)*

*[6 points]* You plot the first score over time, $t_1(t)$ (a score line plot). Describe what each pattern suggests:

(a) a slow monotone increase in $t_1(t)$ over weeks,

(b) a sudden step change in $t_1(t)$ at one time,

(c) a periodic oscillation in $t_1(t)$,

(d) occasional isolated spikes.

For each case, give one typical process/measurement cause.

**Question 3** *(Loading plot: variable relationships)*

*[7 points]* In a loading plot ($p_1$ vs. $p_2$), variables $A$ and $B$ are close together; variable $C$ is roughly opposite them (about 180°); variable $D$ is about 90° away from $A$.

(a) What do these angles imply about pairwise correlations between $(A, B)$, $(A, C)$, and $(A, D)$?

(b) If $A$ and $B$ are far from the origin while $D$ is near the origin, what does that say about their representation in PC1–PC2?

**Question 4** *(Why correlation loading plots are useful even if you have loading plots)*

*[8 points]* You already have a loading plot ($p_1$ vs. $p_2$). Explain *two* reasons why a correlation loading plot can still be useful. Also explain what the 50% and 100% circles mean.

**Question 5** *(Interpreting a correlation loading plot)*

*[7 points]* On a correlation loading plot for PC1–PC2, variable $x_5$ lies inside the 50% circle near the origin. Variable $x_1$ lies close to the outer circle in the positive $t_1$ direction.

(a) What does this imply about how well $x_5$ is modeled by the first two PCs?

(b) What does the position of $x_1$ imply about its relationship with $t_1$ and how it contributes to PC1 interpretation?

(c) Give one action you might take if several important variables lie inside the 50% circle.

**Question 6** *(Biplot reasoning: samples relative to variable arrows)*

*[6 points]* In a biplot (scores + loadings), a sample point lies far in the direction of the arrow for variable "Temperature", and roughly opposite to the arrow for "Yield". What does this say about that sample's Temperature and Yield values relative to the dataset mean?

**Question 7** *($T^2$ vs $Q$ quadrant interpretation)*

*[10 points]* Consider a $T^2$ vs $Q$ plot with critical limits $T^2_{\text{crit}}$ and $Q_{\text{crit}}$ (forming four quadrants). For each quadrant, describe what it typically means and one recommended action:

(a) low $T^2$, low $Q$

(b) high $T^2$, low $Q$

(c) low $T^2$, high $Q$

(d) high $T^2$, high $Q$

**Question 8** *(Influence plot vs $T^2$–$Q$ plot)*

*[7 points]* Some texts use an "influence plot" as *leverage vs residual* while others use $T^2$ vs $Q$.

(a) What is common between these two plots?

(b) What is a key difference in emphasis?

(c) A point has high leverage (or high $T^2$) but low $Q$. Is it automatically a fault? Explain.

**Question 9** *(Why mean centering matters)*

*[7 points]* Explain what can go wrong if you apply PCA to data that is not mean-centered. In particular, what does PC1 tend to capture if the data cloud is far from the origin?

**Question 10** *(Why scaling/normalization matters)*

*[8 points]* You have variables measured in different units (e.g., Temperature in °C, Pressure in bar, Flow in L/min).

(a) What happens if you do PCA on unscaled variables?

(b) When is scaling to unit variance (standardization) recommended?

(c) Give one situation where you might *not* want to scale.

**Question 11** *(Can we apply PCA to categorical variables?)*

*[10 points]* You have a dataset with:

- continuous variables: temperature, pressure
- categorical variables: operator ID (A/B/C), product grade (1/2/3)

(a) Why is "plain PCA" on raw categorical labels not valid?

(b) Name two reasonable approaches for mixed data and state a key caveat for each.

(c) If you one-hot encode categories and run PCA, how do you interpret loadings for those dummy variables?

**Question 12** *(Leave-One-Out (LOO) cross-validation for PCA)*

*[10 points]* You want to choose the number of PCs $r$ using leave-one-out cross-validation (LOO).

(a) Describe the LOO procedure for PCA in words.

(b) A common RMSECV definition is

$$\mathrm{RMSECV}(r) = \sqrt{\frac{\sum_{i=1}^{m} \|\boldsymbol{x}_i - \hat{\boldsymbol{x}}_{i,-i}\|^2}{m \cdot n}}.$$

What are $m$ and $n$ here?

(c) Why can LOO be overly optimistic when samples are strongly correlated (e.g., time series)?

(d) If the RMSECV curve decreases until $r = 4$ and then is nearly flat, what would you choose and why?

**Question 13** *($k$-fold cross-validation for PCA)*

*[10 points]* Compare $k$-fold CV to LOO for choosing the number of PCs.

(a) Describe $k$-fold CV for PCA.

(b) Give two advantages of $k$-fold over LOO in practice.

(c) What is "repeated $k$-fold" and why might it be used?

(d) For time series, what fold design is safer than random folds?

**Question 14** *(Choosing a PCA algorithm)*

*[12 points]* Match each scenario to a suitable PCA computation method and justify briefly:

(a) Small $n$ (e.g., $n = 20$), moderate $m$ (e.g., $m = 500$), no missing values.

(b) Large $m$ (millions), moderate $n$ (hundreds), need the first 5 PCs.

(c) Very large sparse matrix, need approximate first 10 PCs quickly.

(d) Data contains missing values you do not want to impute upfront.

Choose from: eigen-decomposition of covariance, SVD of $\mathbf{X}$, randomized SVD, NIPALS.

**Question 15** *(Building an MSPC model with PCA)*

*[14 points]* Outline a practical workflow to use PCA for process monitoring. Your answer should mention: (i) preprocessing, (ii) model fitting on baseline data, (iii) how to compute monitoring statistics ($T^2$ and $Q$), (iv) how to set control limits, (v) what you do after an alarm.

**Question 16** *(Using moving-window PCA for drift)*

*[8 points]* You suspect the process relationships are slowly changing over months (nonstationary behavior).

(a) What is the idea of moving-window PCA?

(b) What trends in (i) variance explained and (ii) loadings over time indicate drift?

(c) What are two choices for the window step $s$ and what do they mean?

**Question 17** *(Interpreting an alarm: $T^2$ high vs $Q$ high)*

*[8 points]* During monitoring you see three situations:

(a) $T^2$ crosses its limit while $Q$ remains normal,

(b) $Q$ crosses its limit while $T^2$ remains normal,

(c) both cross their limits.

For each, state the most likely interpretation and one diagnostic plot you would use next.